

6(a)	charge / potential (difference)	M1
	charge on one plate, p.d. between the plates	A1
6(b)(i)	all three capacitors connected in series	B1
6(b)(ii)	8 (μF) in parallel with the two 4 (μF) capacitors connected in series	B1
6(c)	discharge from 7.0 V to 4.0 V	C1
	<i>Either</i> energy = $\frac{1}{2}CV^2$ <i>or</i> energy = $\frac{1}{2} QV$ <u>and</u> $C = Q / V$	C1
	energy = $\frac{1}{2} \times 47 \times 10^{-6} \times (7^2 - 4^2)$ = $7.8 \times 10^{-4} \text{ J}$	A1

Question	Answer	Marks
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